9 The densities of three metal alloys, A, B and C, are in the ratio

$$13 : 15 : 21$$

$1\,m^3$ of alloy B has a mass of 8600 kg.

Work out the difference between the mass of $5\,m^3$ of alloy A and $3\,m^3$ of alloy C.
Give your answer correct to 3 significant figures.

1 mark
Density per ratio unit

A	B	C
13	15	21

One ratio unit = 8600 ÷ 15 = 1720/3.

1 mark
Find other densities

Density A = (1720/3) × 13 = 22360/3 kg/m^3.

Density C = (1720/3) × 21 = 12040 kg/m^3.

1 mark
Mass=density×volume

Mass A = (22360/3) × 5 = 111800/3 kg.

Mass C = 12040 × 3 = 36120 kg.

1 mark
Subtract masses

Difference = 111800/3 – 36120 = 3440/3 kg.

1 mark
1150kg

1146.666... kg = 1150 kg (3 significant figures).

..... kg

(Total for Question 9 is 5 marks)